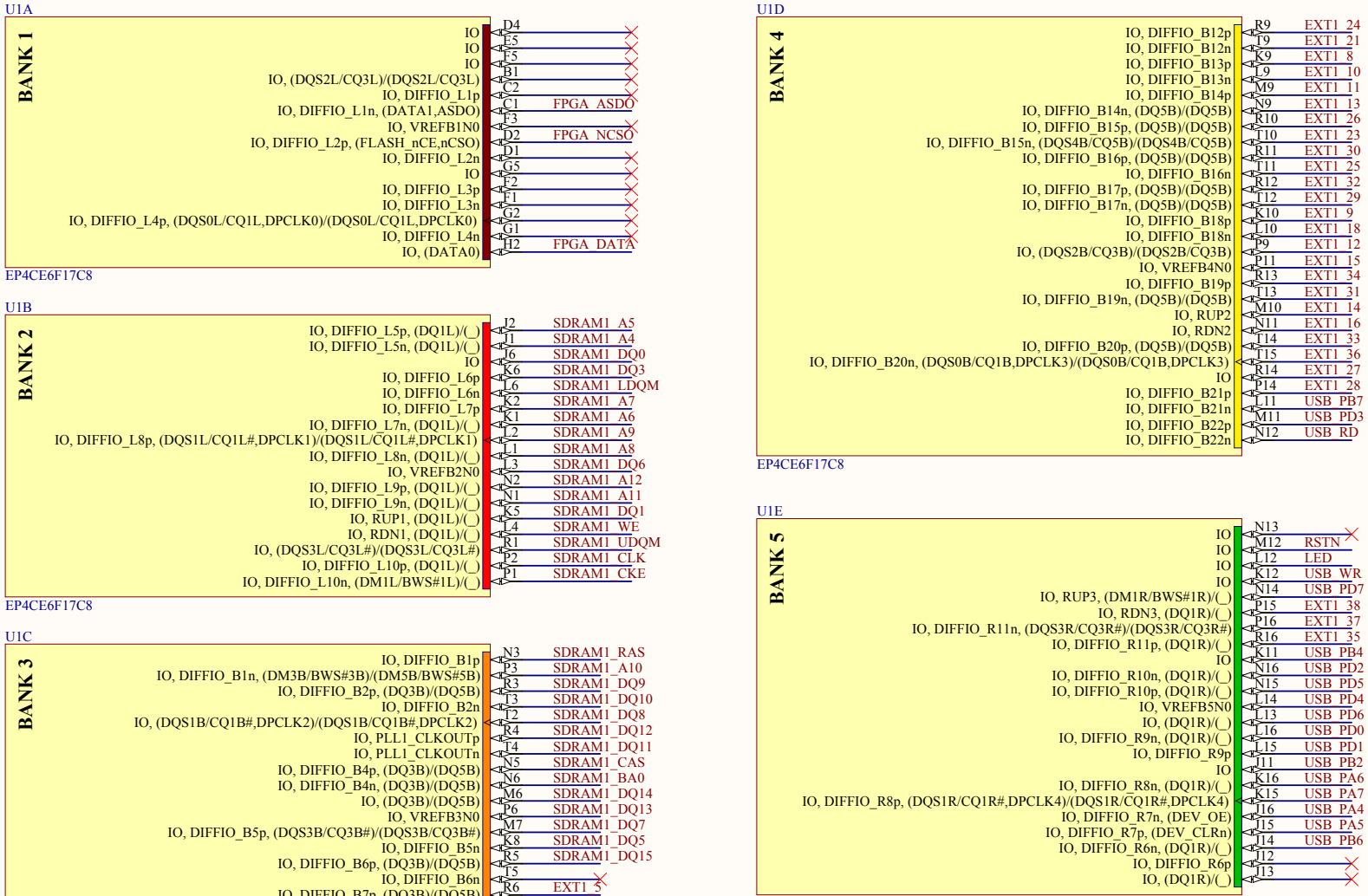
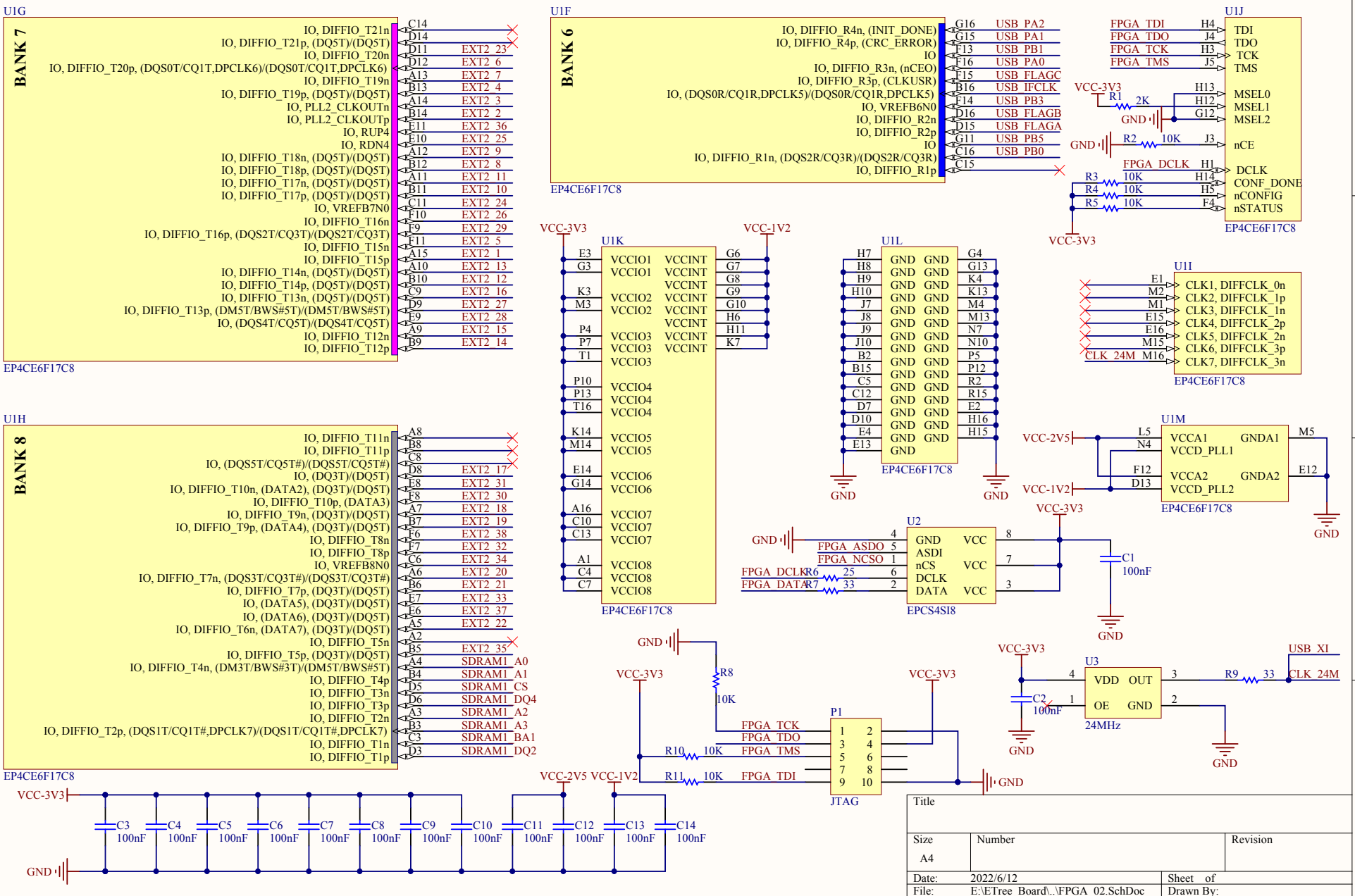
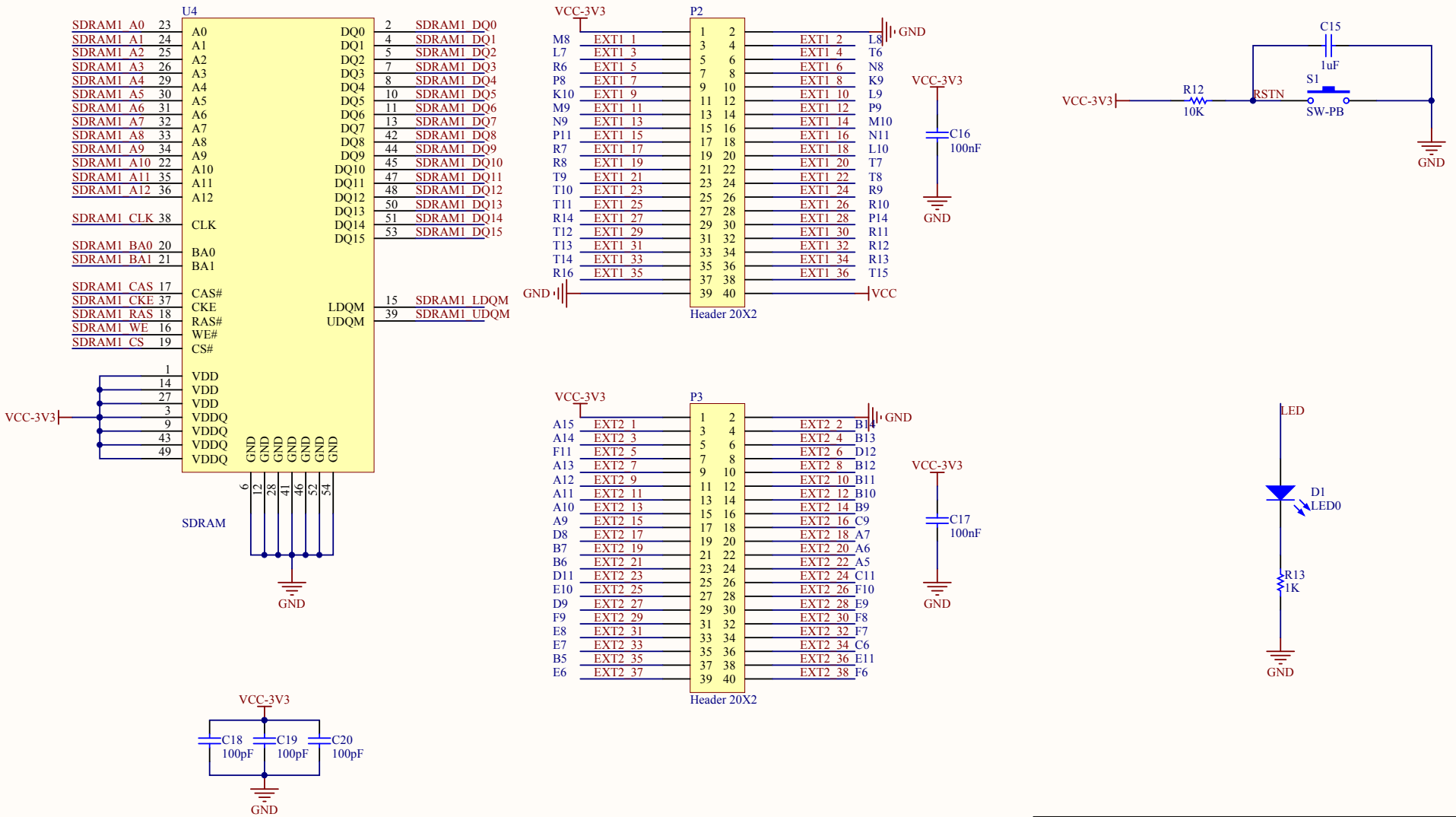
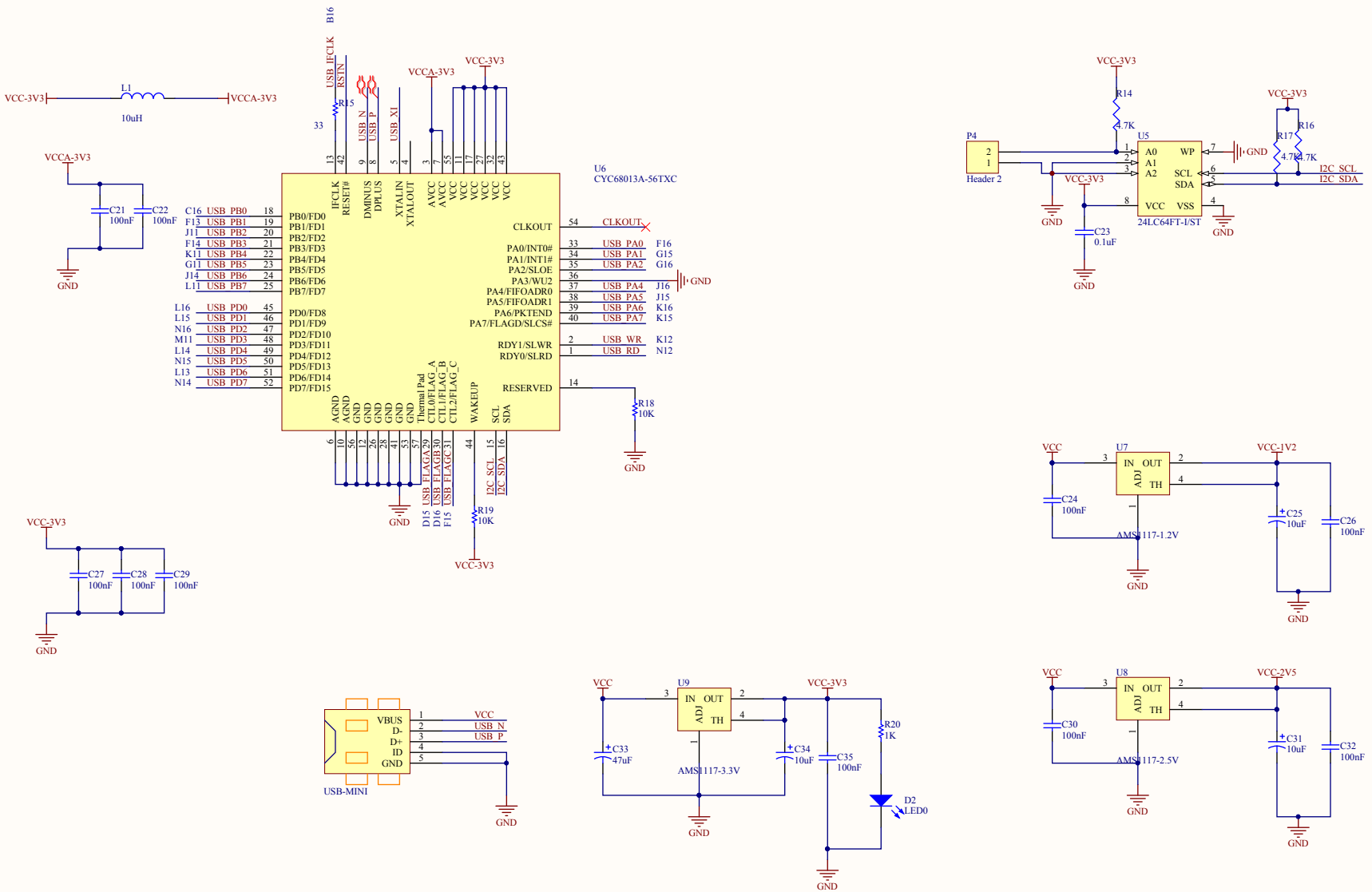

U1A
BANK 1
IO
IO
IO
IO, (DQS2L/CQ3L)/(DQS2L/CQ3L)
IO, DIFFIO_L1p
IO, DIFFIO_L1n, (DATA1,ASDO)
IO, VREFB1N0
IO, DIFFIO_L2p, (FLASH_nCE,nCSO)
IO, DIFFIO_L2n
IO
IO, DIFFIO_L3p
IO, DIFFIO_L3n
IO, DIFFIO_L4p, (DQS0L/CQ1L,DPCLK0)/(DQS0L/CQ1L,DPCLK0)
IO, DIFFIO_L4n
IO, (DATA0)
D4
E5
F5
B1
C2
C1
F3
D2
D1
G5
F2
F1
G2
G1
H2
FPGA_ASDO
FPGA_NCSO
FPGA_DATA
EP4CE6F17C8
U1B
BANK 2
IO, DIFFIO_L5p, (DQ1L)/(_)
IO, DIFFIO_L5n, (DQ1L)/(_)
IO
IO, DIFFIO_L6p
IO, DIFFIO_L6n
IO, DIFFIO_L7p
IO, DIFFIO_L7n, (DQ1L)/(_)
IO, DIFFIO_L8p, (DQS1L/CQ1L#,DPCLK1)/(DQS1L/CQ1L#,DPCLK1)
IO, DIFFIO_L8n, (DQ1L)/(_)
IO, VREFB2N0
IO, DIFFIO_L9p, (DQ1L)/(_)
IO, DIFFIO_L9n, (DQ1L)/(_)
IO, RUP1, (DQ1L)/(_)
IO, RDN1, (DQ1L)/(_)
IO, (DQS3L/CQ3L#)/(DQS3L/CQ3L#)
IO, DIFFIO_L10p, (DQ1L)/(_)
IO, DIFFIO_L10n, (DM1L/BWS#1L)/(_)
J2
J1
J6
K6
L6
K2
K1
L2
L1
L3
N2
N1
K5
L4
R1
P2
P1
SDRAM1_A5
SDRAM1_A4
SDRAM1_DQ0
SDRAM1_DQ3
SDRAM1_LDQM
SDRAM1_A7
SDRAM1_A6
SDRAM1_A9
SDRAM1_A8
SDRAM1_DQ6
SDRAM1_A12
SDRAM1_A11
SDRAM1_DQ1
SDRAM1_WE
SDRAM1_UDQM
SDRAM1_CLK
SDRAM1_CKE
EP4CE6F17C8
U1C
BANK 3
IO, DIFFIO_B1p
IO, DIFFIO_B1n, (DM3B/BWS#3B)/(DM5B/BWS#5B)
IO, DIFFIO_B2p, (DQ3B)/(DQ5B)
IO, DIFFIO_B2n
IO, (DQS1B/CQ1B#,DPCLK2)/(DQS1B/CQ1B#,DPCLK2)
IO, PLL1_CLKOUTp
IO, PLL1_CLKOUTn
IO, DIFFIO_B4p, (DQ3B)/(DQ5B)
IO, DIFFIO_B4n, (DQ3B)/(DQ5B)
IO, (DQ3B)/(DQ5B)
IO, VREFB3N0
IO, DIFFIO_B5p, (DQS3B/CQ3B#)/(DQS3B/CQ3B#)
IO, DIFFIO_B5n
IO, DIFFIO_B6p, (DQ3B)/(DQ5B)
IO, DIFFIO_B6n
IO, DIFFIO_B7p, (DQ3B)/(DQ5B)
IO, DIFFIO_B7n
IO, (DQ3B)/(DQ5B)
IO, DIFFIO_B8p, (DQ3B)/(DQ5B)
IO, DIFFIO_B8n, (DQS5B/CQ5B#)/(DQS5B/CQ5B#)
IO, DIFFIO_B9p, (DQ3B)/(DQ5B)
IO, DIFFIO_B9n, (DM5B/BWS#5B)/(DM5B/BWS#5B)
IO, DIFFIO_B10p, (DQ5B)/(DQ5B)
IO, DIFFIO_B10n, (DQ5B)/(DQ5B)
IO, DIFFIO_B11p
IO, DIFFIO_B11n
N3
P3
R3
T3
T2
R4
T4
N5
N6
M6
P6
M7
K8
K5
T5
R6
T6
L7
R7
T7
L8
M8
N8
P8
R8
T8
SDRAM1_RAS
SDRAM1_A10
SDRAM1_DQ9
SDRAM1_DQ10
SDRAM1_DQ8
SDRAM1_DQ12
SDRAM1_DQ11
SDRAM1_CAS
SDRAM1_BA0
SDRAM1_DQ14
SDRAM1_DQ13
SDRAM1_DQ7
SDRAM1_DQ5
SDRAM1_DQ15
EXT1_5
EXT1_4
EXT1_3
EXT1_17
EXT1_20
EXT1_2
EXT1_1
EXT1_6
EXT1_7
EXT1_19
EXT1_22
EP4CE6F17C8
U1D
BANK 4
IO, DIFFIO_B12p
IO, DIFFIO_B12n
IO, DIFFIO_B13p
IO, DIFFIO_B13n
IO, DIFFIO_B14p
IO, DIFFIO_B14n, (DQ5B)/(DQ5B)
IO, DIFFIO_B15p, (DQ5B)/(DQ5B)
IO, DIFFIO_B15n, (DQS4B/CQ5B)/(DQS4B/CQ5B)
IO, DIFFIO_B16p, (DQ5B)/(DQ5B)
IO, DIFFIO_B16n
IO, DIFFIO_B17p, (DQ5B)/(DQ5B)
IO, DIFFIO_B17n, (DQ5B)/(DQ5B)
IO, DIFFIO_B18p
IO, DIFFIO_B18n
IO, (DQS2B/CQ3B)/(DQS2B/CQ3B)
IO, VREFB4N0
IO, DIFFIO_B19p
IO, DIFFIO_B19n, (DQ5B)/(DQ5B)
IO, RUP2
IO, RDN2
IO, DIFFIO_B20p, (DQ5B)/(DQ5B)
IO, DIFFIO_B20n, (DQS0B/CQ1B,DPCLK3)/(DQS0B/CQ1B,DPCLK3)
IO
IO, DIFFIO_B21p
IO, DIFFIO_B21n
IO, DIFFIO_B22p
IO, DIFFIO_B22n
R9
T9
K9
L9
M9
V9
R10
T10
R11
T11
R12
T12
K10
L10
P9
P11
R13
T13
M10
N11
T14
T15
R14
P14
L11
M11
N12
EXT1_24
EXT1_21
EXT1_8
EXT1_10
EXT1_11
EXT1_13
EXT1_26
EXT1_23
EXT1_30
EXT1_25
EXT1_32
EXT1_29
EXT1_9
EXT1_18
EXT1_12
EXT1_15
EXT1_34
EXT1_31
EXT1_14
EXT1_16
EXT1_33
EXT1_36
EXT1_27
EXT1_28
USB_PB7
USB_PD3
USB_RD
EP4CE6F17C8
U1E
BANK 5
IO
IO
IO
IO
IO, RUP3, (DM1R/BWS#1R)/(_)
IO, RDN3, (DQ1R)/(_)
IO, DIFFIO_R11n, (DQS3R/CQ3R#)/(DQS3R/CQ3R#)
IO, DIFFIO_R11p, (DQ1R)/(_)
IO
IO, DIFFIO_R10n, (DQ1R)/(_)
IO, DIFFIO_R10p, (DQ1R)/(_)
IO, VREFB5N0
IO, (DQ1R)/(_)
IO, DIFFIO_R9n, (DQ1R)/(_)
IO, DIFFIO_R9p
IO
IO, DIFFIO_R8n, (DQ1R)/(_)
IO, DIFFIO_R8p, (DQS1R/CQ1R#,DPCLK4)/(DQS1R/CQ1R#,DPCLK4)
IO, DIFFIO_R7n, (DEV_OE)
IO, DIFFIO_R7p, (DEV_CLRn)
IO, DIFFIO_R6n, (DQ1R)/(_)
IO, DIFFIO_R6p
IO, (DQ1R)/(_)
N13
M12
L12
K12
N14
P15
P16
R16
K11
N16
N15
L14
L13
L16
L15
J11
K16
K15
J16
J15
J14
J12
J13
RSTN
LED
USB_WR
USB_PD7
EXT1_38
EXT1_37
EXT1_35
USB_PB4
USB_PD2
USB_PD5
USB_PD4
USB_PD6
USB_PD0
USB_PD1
USB_PB2
USB_PA6
USB_PA7
USB_PA4
USB_PA5
USB_PB6
EP4CE6F17C8
Title
Size
A4
Number
Revision
Date:
2022/6/12
Sheet of
File:
E:\ETree Board\..\FPGA_01.SchDoc
Drawn By:


U1G
BANK 7
IO, DIFFIO_T21n
IO, DIFFIO_T21p, (DQ5T)/(DQ5T)
IO, DIFFIO_T20n
IO, DIFFIO_T20p, (DQS0T/CQ1T,DPCLK6)/(DQS0T/CQ1T,DPCLK6)
IO, DIFFIO_T19n
IO, DIFFIO_T19p, (DQ5T)/(DQ5T)
IO, PLL2_CLKOUTn
IO, PLL2_CLKOUTp
IO, RUP4
IO, RDN4
IO, DIFFIO_T18n, (DQ5T)/(DQ5T)
IO, DIFFIO_T18p, (DQ5T)/(DQ5T)
IO, DIFFIO_T17n, (DQ5T)/(DQ5T)
IO, DIFFIO_T17p, (DQ5T)/(DQ5T)
IO, VREFB7N0
IO, DIFFIO_T16n
IO, DIFFIO_T16p, (DQS2T/CQ3T)/(DQS2T/CQ3T)
IO, DIFFIO_T15n
IO, DIFFIO_T15p
IO, DIFFIO_T14n, (DQ5T)/(DQ5T)
IO, DIFFIO_T14p, (DQ5T)/(DQ5T)
IO, DIFFIO_T13n, (DQ5T)/(DQ5T)
IO, DIFFIO_T13p, (DM5T/BWS#5T)/(DM5T/BWS#5T)
IO, (DQS4T/CQ5T)/(DQS4T/CQ5T)
IO, DIFFIO_T12n
IO, DIFFIO_T12p
C14
D14
D11 EXT2_23
D12 EXT2_6
A13 EXT2_7
B13 EXT2_4
A14 EXT2_3
B14 EXT2_2
E11 EXT2_36
E10 EXT2_25
A12 EXT2_9
B12 EXT2_8
A11 EXT2_11
B11 EXT2_10
C11 EXT2_24
F10 EXT2_26
F9 EXT2_29
F11 EXT2_5
A15 EXT2_1
A10 EXT2_13
B10 EXT2_12
C9 EXT2_16
D9 EXT2_27
E9 EXT2_28
A9 EXT2_15
B9 EXT2_14
EP4CE6F17C8
U1F
BANK 6
IO, DIFFIO_R4n, (INIT_DONE)
IO, DIFFIO_R4p, (CRC_ERROR)
IO
IO, DIFFIO_R3n, (nCEO)
IO, DIFFIO_R3p, (CLKUSR)
IO, (DQS0R/CQ1R,DPCLK5)/(DQS0R/CQ1R,DPCLK5)
IO, VREFB6N0
IO, DIFFIO_R2n
IO, DIFFIO_R2p
IO
IO, DIFFIO_R1n, (DQS2R/CQ3R)/(DQS2R/CQ3R)
IO, DIFFIO_R1p
G16 USB_PA2
G15 USB_PA1
F13 USB_PB1
F16 USB_PA0
F15 USB_FLAGC
B16 USB_IFCLK
F14 USB_PB3
D16 USB_FLAGB
D15 USB_FLAGA
G11 USB_PB5
C16 USB_PB0
C15
EP4CE6F17C8
U1J
FPGA_TDI H4 TDI
FPGA_TDO J4 TDO
FPGA_TCK H3 TCK
FPGA_TMS J5 TMS
H13 MSEL0
H12 MSEL1
G12 MSEL2
J3 nCE
FPGA_DCLK H1 DCLK
H14 CONF_DONE
H5 nCONFIG
F4 nSTATUS
VCC-3V3
R1 2K
GND
R2 10K
R3 10K
R4 10K
R5 10K
EP4CE6F17C8
U1K
VCCIO1
VCCIO2
VCCIO3
VCCIO4
VCCIO5
VCCIO6
VCCIO7
VCCIO8
VCCINT
VCC-3V3
VCC-1V2
EP4CE6F17C8
U1L
GND
EP4CE6F17C8
U1I
CLK1, DIFFCLK_0n
CLK2, DIFFCLK_1p
CLK3, DIFFCLK_1n
CLK4, DIFFCLK_2p
CLK5, DIFFCLK_2n
CLK6, DIFFCLK_3p
CLK7, DIFFCLK_3n
CLK_24M M16
EP4CE6F17C8
U1H
BANK 8
IO, DIFFIO_T11n
IO, DIFFIO_T11p
IO, (DQS5T/CQ5T#)/(DQS5T/CQ5T#)
IO, (DQ3T)/(DQ5T)
IO, DIFFIO_T10n, (DATA2), (DQ3T)/(DQ5T)
IO, DIFFIO_T10p, (DATA3)
IO, DIFFIO_T9n, (DQ3T)/(DQ5T)
IO, DIFFIO_T9p, (DATA4), (DQ3T)/(DQ5T)
IO, DIFFIO_T8n
IO, DIFFIO_T8p
IO, VREFB8N0
IO, DIFFIO_T7n, (DQS3T/CQ3T#)/(DQS3T/CQ3T#)
IO, DIFFIO_T7p, (DQ3T)/(DQ5T)
IO, (DATA5), (DQ3T)/(DQ5T)
IO, (DATA6), (DQ3T)/(DQ5T)
IO, DIFFIO_T6n, (DATA7), (DQ3T)/(DQ5T)
IO, DIFFIO_T5n
IO, DIFFIO_T5p, (DQ3T)/(DQ5T)
IO, DIFFIO_T4n, (DM3T/BWS#3T)/(DM5T/BWS#5T)
IO, DIFFIO_T4p
IO, DIFFIO_T3n
IO, DIFFIO_T3p
IO, DIFFIO_T2n
IO, DIFFIO_T2p, (DQS1T/CQ1T#,DPCLK7)/(DQS1T/CQ1T#,DPCLK7)
IO, DIFFIO_T1n
IO, DIFFIO_T1p
EXT2_17
EXT2_31
EXT2_30
EXT2_18
EXT2_19
EXT2_38
EXT2_32
EXT2_34
EXT2_20
EXT2_21
EXT2_33
EXT2_37
EXT2_22
EXT2_35
SDRAM1_A0
SDRAM1_A1
SDRAM1_CS
SDRAM1_DQ4
SDRAM1_A2
SDRAM1_A3
SDRAM1_BA1
SDRAM1_DQ2
EP4CE6F17C8
U1M
VCCA1
VCCA2
GNDA1
GNDA2
VCCD_PLL1
VCCD_PLL2
VCC-2V5
VCC-1V2
EP4CE6F17C8
U2
GND
ASDI
nCS
DCLK
DATA
VCC
FPGA_ASDO
FPGA_NCSO
FPGA_DCLK
FPGA_DATA
R6 25
R7 33
EPCS4SI8
C1 100nF
P1
JTAG
FPGA_TCK
FPGA_TDO
FPGA_TMS
FPGA_TDI
R8 10K
R10 10K
R11 10K
U3
VDD
OUT
OE
24MHz
C2 100nF
R9 33
USB_XI
CLK_24M
C3 100nF
C4 100nF
C5 100nF
C6 100nF
C7 100nF
C8 100nF
C9 100nF
C10 100nF
C11 100nF
C12 100nF
C13 100nF
C14 100nF
Title
Size A4
Number
Revision
Date: 2022/6/12
Sheet of
File: E:\ETree_Board\..\FPGA_02.SchDoc
Drawn By:



Title		
Size A4	Number	Revision
Date: 2022/6/12		Sheet of
File: E:\ETree_Board\..\SDRAM.SchDoc		Drawn By:



Title		
Size A3	Number	Revision
Date: 2022/6/12		Sheet of
File: E:\ETree_Board\..\USB2.SchDoc		Drawn By: